Quiz6 - Results

Attempt 1 of 1

Written Mar 10, 2026 5:01 PM - Mar 10, 2026 5:07 PM

Attempt Score	**10 / 10 - 100 %**
Overall Grade (Highest Attempt)	**10 / 10 - 100 %**

Question 1 1 / 1 point

Why do Vision Transformers divide an input image into patches rather than processing each individual pixel independently?

- ○ The patches ensure more stable training of the model, helping the loss decrease faster
- ○ The patches help to separate the input image into completely independent red, green, and blue sequences.
- ○ The patches act as the positional embedding for the transformer
- ✓ ◉ The patches help to learn from spatial neighborhoods, avoiding the assumption that pixels contribute independently.

Question 2 1 / 1 point

You are training a model for document summarization. The BLEU score you calculate on the test set will be of better quality if you use one reference for each test document, rather than multiple references.

- ○ True
- ✓ ◉ False

Question 3 1 / 1 point

Typically, fine-tuning involves a much larger amount of training data than pre-training.

- ○ True
- ✓ ◉ False

Question 4 1 / 1 point

Which of the following sequences of training an LLM is most typical?

- ✓ ◉ Pre-training –> Fine-tuning –> RL training
- ○ Fine-tuning –> Pre-training –> RL training
- ○ Pre-training –> RL training –> Fine-tuning
- ○ Pre-training –> Fine-tuning –> Continue pre-training –> RL training

Question 5 1 / 1 point

LLMs experience “exposure bias” after the fine-tuning step. Why does this happen?

- ○ The LLM has never before predicted the next word based on its own past generations
- ○ The LLM has so far only predicted the next word based on the given human-written prefix
- ✓ ◉ Both of the above
- ○ None of the above

Question 6 1 / 1 point

Which of the following NLP tasks will benefit from diversity in generated outputs?

- ○ "Predict the language this sentence is written in."
- ✓ ◉ "Tell me a joke!"
- ○ Named Entity Recognition
- ○ POS Tagging

Question 7 — 1 / 1 point

Sahana is training an LLM to solve math problems. A numerical score that represents the logical correctness of the reasoning generated by the LLM can be used as a verifiable reward for RL training.

- ✓ (•) True
- ( ) False

Question 8 — 1 / 1 point

Which of the following represents an "action" that can be taken by the RL agent when applying RL to NLP?

- ( ) Assess the correctness of its generation so far
- ✓ (•) Generate the next token
- ( ) Backspace the previously generated token
- ( ) All of the above

Question 9 — 1 / 1 point

Which of the following statements are true about Vision-and-Language Models (VLM) like Qwen3-VL? Choose all that apply.

- ✓ [ ] It is a type of late fusion model
- ✓ [x] Its architecture includes a vision transformer (ViT) to process image inputs
- ✓ [x] It can be used for visual question answering
- ✓ [x] It can be used for automatic image captioning

Question 10 — 1 / 1 point

Sahana trained a sentiment analysis classifier that has to predict if a given sentence is happy or sad. On an imbalanced test dataset, she gets the following results:

"Happy" class: True Positives (TP) = 80, False Positives (FP) = 20

"Sad" class: True Positives (TP) = 10, False Positives (FP) = 40

What are the macro-average and micro-average precision for these results? Note that precision = TP / (TP + FP).

- ✔ (selected) Macro: 0.50 | Micro: 0.60
- Macro: 0.50 | Micro: 0.50
- Macro: 0.45 | Micro: 0.55
- Macro: 0.60 | Micro: 0.50

Done